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B.Tech. Degree V Semester Examination November 2014

IT 1504 DESIGN AND ANALYSIS OF ALGORITHMS (2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) What are the different asymptotic notations used to compare different algorithms?
 (b) How can recursive procedures be implemented using activation frames?
 (c) Write an algorithm for merge sort and analyze its complexity.
 (d) List out the properties of Red Black tree.
 (e) Write short note on strongly connected component of a graph.
 (f) Explain any one graph traversal algorithm.
 (g) Explain Widgerson's algorithm for graph coloring.
 (h) Write short note on approximation algorithms.



PART B

(4 × 15 = 60)

- II. (a) Explain different criteria to analyze an algorithm to find the running time. (10)
 (b) Solve the following recurrence equation (5)
 $T(n) = 2T(\sqrt{n}) + \log n$.
- OR**
- III. What is meant by dynamic programming? How to find an optimal parenthesization of matrix-chain product using dynamic programming? (15)
- IV. Explain heap sort procedure with an example. Analyze its complexity. (15)
- OR**
- V. (a) Write an algorithm for insertion sorting method and analyze its complexity. (8)
 (b) Explain binomial heap with an example. (7)
- VI. (a) Discuss any two methods to represent a graph. (5)
 (b) Explain the algorithm to find all pair shortest path in a graph with the help of an example. (10)
- OR**
- VII. (a) Explain Warshall's algorithm to find transitive closure of a binary relation. (7)
 (b) Explain the procedure for constructing optimal binary search trees. (8)
- VIII. (a) Explain various strategies applied to Travelling Salesman Problem. (10)
 (b) Write short note on NP-completeness. (5)
- OR**
- IX. Explain various strategies for Bin-packing problem. (15)